

DEPARTMENT OF PHYSICS
INDIAN INSTITUTE OF TECHNOLOGY, MADRAS
PH1020 Physics II
Lecture Note-3

Energy of dipole Date: 5 Feb 2025

1. Problem 3.36: Electric field of a perfect dipole.

Question: Show that the electric field of a perfect dipole can be written in the *coordinate free form* as

$$\boxed{\boldsymbol{E}_{dip}(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\frac{1}{r^3}[3(\boldsymbol{p}\cdot\hat{\boldsymbol{r}})\hat{\boldsymbol{r}} - \boldsymbol{p}]}$$

Solution:

Electric field of a perfect dipole $\boldsymbol{p}$ which is placed at the origin and points in z-direction is given by Griffiths' book Eq.[3.103]

$$\boldsymbol{E}_{\text{dip}}\,(r,\theta) = \frac{p}{4\pi\epsilon_0 r^3}(2\cos\theta\hat{\boldsymbol{r}} + \sin\theta\widehat{\boldsymbol{\theta}})$$
$$= \frac{1}{4\pi\epsilon_0 r^3}(3p\cos\theta\hat{\boldsymbol{r}} - p\cos\theta\hat{\boldsymbol{r}} + p\sin\theta\widehat{\boldsymbol{\theta}})$$

We can write

$$p\cos\theta = \boldsymbol{p}\cdot\hat{\boldsymbol{r}}$$
$$p\sin\theta = -\boldsymbol{p}\cdot\widehat{\boldsymbol{\theta}}$$
$$\boldsymbol{E}_{\text{dip}}\,(r,\theta) = \frac{1}{4\pi\epsilon_0 r^3}[3(\boldsymbol{p}\cdot\hat{\boldsymbol{r}})\hat{\boldsymbol{r}} - (\boldsymbol{p}\cdot\hat{\boldsymbol{r}})\hat{\boldsymbol{r}} - (\boldsymbol{p}\cdot\widehat{\boldsymbol{\theta}})\widehat{\boldsymbol{\theta}}]$$

Since

$$\boldsymbol{p} = (\boldsymbol{p}\cdot\hat{\boldsymbol{r}})\hat{\boldsymbol{r}} + (\boldsymbol{p}\cdot\widehat{\boldsymbol{\theta}})\widehat{\boldsymbol{\theta}}$$

We obtain

$$\boldsymbol{E}_{\text{dip}}\,(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\frac{1}{r^3}[3(\boldsymbol{p}\cdot\hat{\boldsymbol{r}})\hat{\boldsymbol{r}} - \boldsymbol{p}] \tag{1}$$

2. Problem 4.7: Energy of an ideal dipole $\boldsymbol{p}$ in an electric field $\boldsymbol{E}$

Question:

Show that the energy of an ideal dipole $\boldsymbol{p}$ in an electric field $\boldsymbol{E}$ is given by

$$\boxed{U = -\boldsymbol{p}\cdot\boldsymbol{E}}$$

Solution:

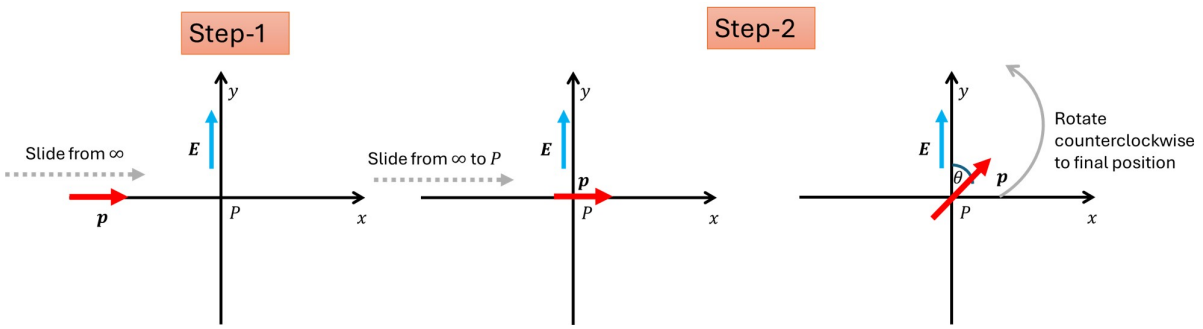


Figure 1: Steps to bring the dipole $\boldsymbol{p}$ from infinity in to the electric field $\boldsymbol{E}$.

Without any loss of generality we can assume the field $\boldsymbol{E}$ is uniform and points in the $\boldsymbol{y}$-direction.

We know that work done is path independent. Lets take one particular path : first slide $\boldsymbol{p}$ in from infinity along the x-axis up to P (Step-1) then rotate the dipole counter-clockwise into the final position (Step-2) as shown in Fig. 1

Work done in Step-1 is given by

$$U_1 = \int \boldsymbol{F} \cdot d\boldsymbol{l} = 0$$

This is because $\boldsymbol{F} \perp d\boldsymbol{l}$ throughout the path.

In the step-2 we rotate the dipole counterclockwise into the final position. The torque exerted by $\boldsymbol{E}$ on the dipole is

$$\begin{aligned} \boldsymbol{N} &= \boldsymbol{p} \times \boldsymbol{E} \\ &= pE \sin\theta \hat{\mathbf{z}} \end{aligned}$$

Torque we have to exert $= pE \sin\theta \hat{\mathbf{z}}$ clockwise

$$\begin{aligned} U_2 &= \int_{\pi/2}^{\theta} pE \sin\theta' d\theta' \\ &= pE \left(-\cos\theta'\right)\Big|_{\frac{\pi}{2}}^{\theta} \\ &= -pE \left(\cos\theta - \cos\frac{\pi}{2}\right) \\ &= -pE\cos\theta \end{aligned}$$

Therefore total work done or energy of the dipole is given by $U = U_1 + U_2$

$$U = -\boldsymbol{p} \cdot \boldsymbol{E} \tag{2}$$

3. Problem 4.8: The interaction energy between two dipoles.

Question: Show that the interaction energy of two dipoles separated by a displacement $\boldsymbol{r}$ is

$$\boxed{U = \frac{1}{4\pi\epsilon_0}\frac{1}{r^3}\left[\boldsymbol{p}_1 \cdot \boldsymbol{p}_2 - 3\left(\boldsymbol{p}_1 \cdot \hat{\boldsymbol{r}}\right)\left(\boldsymbol{p}_2 \cdot \hat{\boldsymbol{r}}\right)\right]}$$

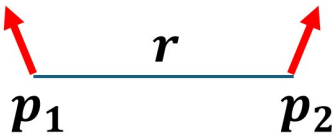


Figure 2: Two dipoles separated by a displacement $\boldsymbol{r}$.

Solution:

From Eq. 2 we know that the energy of the dipole $\boldsymbol{p_1}$ due to field $\boldsymbol{E_2}$

$$U = -\boldsymbol{p_1} \cdot \boldsymbol{E_2}$$

Electric field due to dipole $\boldsymbol{p_2}$ is given by Eq. 1

$$\boldsymbol{E}_2 = \frac{1}{4\pi\epsilon_0}\frac{1}{r^3}\left[3\left(\boldsymbol{p}_2 \cdot \hat{r}\right)\hat{\boldsymbol{r}} - \boldsymbol{p}_2\right]$$

Therefore

$$U = -\boldsymbol{p}_1 \cdot \frac{1}{4\pi\epsilon_0}\frac{1}{r^3}\left[3\left(\boldsymbol{p}_2 \cdot \hat{r}\right)\hat{\boldsymbol{r}} - \boldsymbol{p}_2\right]$$

$$= \frac{1}{4\pi\epsilon_0}\frac{1}{r^3}\left[\boldsymbol{p}_1 \cdot \boldsymbol{p}_2 - 3\left(\boldsymbol{p}_1 \cdot \hat{\boldsymbol{r}}\right)\left(\boldsymbol{p}_2 \cdot \hat{\boldsymbol{r}}\right)\right]$$

$$\boxed{U_{12} = U_{21}}$$

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